Dr. Atul Suresh Deshpande
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Department of Materials Science & Metallurgical Engineering
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Education

- **Ph.D :** September 2001-June 2004, International Max Planck Research School (IMPRS) on Biomimetic Systems, Max Planck Institute of Colloids and interfaces, Potsdam, Germany.
Thesis title: Fabrication of porous metal oxides for catalytic applications using templating techniques
- **M.Sc. (Inorganic Chemistry):** August1997- June1999: Department of Chemistry, University of Pune, India
- **B.Sc. (Chemistry)** August1994- June 1997 Sir Parashurambhau College, Pune, India.

Professional positions

December 2019- till date: Associate Professor at Department of Materials Science and Metallurgical Engineering, Indian Institute of Technology Hyderabad, Hyderabad, India
December 2011- December 2019: Assistant Professor at Department of Materials Science and Metallurgical Engineering, Indian Institute of Technology Hyderabad, Hyderabad, India
August 2006- February 2011: Postdoctoral Fellow at the Forsyth Institute, Boston, US, till August 2007. Continued the same position at the School of Dental Medicine, University of Pittsburgh, Pittsburgh, US.
August 2004- April 2006: Postdoctoral fellow at the Department of Biomaterials, Max Planck Institute of Colloids and interfaces, Potsdam, Germany.
July1999-July 2001: Project assistant, the Physical and Materials Chemistry Division, National Chemical Laboratory, Pune.

Research Interests

Nanoparticle synthesis: Synthesis of novel multicomponent oxide (high entropy oxide) and high entropy alloy nanoparticles, their detailed structural analysis and analysis of their functional properties.
Carbon Materials: Generation of nanostructured carbon materials for energy storage applications
Biomaterials: Biomimetic approaches for the generation of materials for superhydrophobic surfaces and coatings

Number of PhD students (graduated): 5
Number of PhD students (ongoing): 3

List of Publications and Patents:
Total number of Publications: 48; Patents: 3; H-index: 28

List of Publications:

1. Pore-Engineered Ultrahigh Surface Area Okra-Derived Activated Carbon for Concurrently High-Power-Density and High-Energy-Density Supercapacitor.Majhi E, Deshpande AS. ***Batteries and Supercaps***. Published online 2025 (**2025**). doi:10.1002/batt.202500160

2. Quantum capacitance: The large but hidden capacitance in supercapacitors.Kumar S, Majhi E, Deshpande AS, Khandelwal M. ***Carbon Trends***. 16:100385, (**2024**). doi:https://doi.org/10.1016/j.cartre.2024.100385

3. High temperature elemental segregation induced structure degradation in high entropy fluorite oxide.Hu Y, Anandkumar M, Zhang Y, et al. ***Journal of Advanced Ceramics***. Published online 2024 (**2024**). doi:10.26599/JAC.2024.9220854

4. Improved chemiresistor gas sensing response by optimizing the applied electric field and interdigitated electrode geometry.Naganaboina VR, Bonam S, Anandkumar M, Deshpande AS, Singh SG. ***Materials Chemistry and Physics***. 305 (**2023**). doi:10.1016/j.matchemphys.2023.127975

5. PEDOT:PSS-bacterial cellulose bilayer actuators: From the movement of ions to deflection.Najathulla BC, Kumar S, Deshpande AS, Khandelwal M. ***Polymers for Advanced Technologies***. Published online 2023 (**2023**). doi:10.1002/pat.6040

6. Effective band gap engineering in multi-principal oxides (CeGdLa-Zr/Hf)Ox by temperature-induced oxygen vacancies.Hu Y, Anandkumar M, Joardar J, Wang X, Deshpande AS, Reddy KM. ***Scientific Reports***. 13(1):2362, (**2023**). doi:10.1038/s41598-023-29477-0

7. TiO2 Decorated SiO2 Nanoparticles as Efficient Antibacterial Materials: Enhanced Activity under Low Power UV Light.Mahanta U, Deshpande AS, Khandelwal M. ***ChemistrySelect***. 8(4):e202203724, (**2023**). doi:https://doi.org/10.1002/slct.202203724

8. Biocompatible and antimicrobial multilayer fibrous polymeric wound dressing with optimally embedded silver nanoparticles.Sarviya N, Mahanta U, Dart A, et al. ***Applied Surface Science***. 612:155799, (**2023**). doi:https://doi.org/10.1016/j.apsusc.2022.155799

9. Humidity-Independent Methane Gas Detection in Gd0.2La0.2Ce0.2Hf0.2Zr0.2O2-based Sensor Using Polynomial Regression Analysis.Naganaboina VR, Bonam S, Anandkumar M, Deshpande AS, Singh SG. ***IEEE Electron Device Letters***. Published online 2022:1, (**2022**). doi:10.1109/LED.2022.3215616

10. Smartphone camera-based micron-scale displacement measurement: development and

application in soft actuators.Najathulla BC, Deshpande AS, Khandelwal M. ***Instrumentation Science & Technology***. Published online March 22, 2022:1-10, (**2022**). doi:10.1080/10739149.2022.2053153

11. Single-Phase High-Entropy Oxide Nanoparticles for Wide Dynamic Range Detection of CO 2.Naganaboina VR, Anandkumar M, Deshpande AS, Singh SG. ***ACS Applied Nano Materials***. 5(3):4524-4536, (**2022**). doi:10.1021/acsanm.2c00855

12. PEDOT:PSS/Bacterial Cellulose-based soft actuator under triangle and square wave: Deflection, response and fidelity.Najathulla BC, Deshpande AS, Khandelwal M. ***Synthetic Metals***. 286:117053, (**2022**). doi:10.1016/j.synthmet.2022.117053

13. Single-phase high-entropy oxide-based chemiresistor: Toward selective and sensitive detection of methane gas for real-time applications.Naganaboina VR, Anandkumar M, Deshpande AS, Singh SG. ***Sensors and Actuators B: Chemical***. 357:131426, (**2022**). doi:10.1016/j.snb.2022.131426

14. TiO2@SiO2 nanoparticles for methylene blue removal and photocatalytic degradation under natural sunlight and low-power UV light.Mahanta U, Khandelwal M, Deshpande AS. ***Applied Surface Science***. 576:151745, (**2022**). doi:10.1016/j.apsusc.2021.151745

15. Sweetcorn husk derived porous carbon with inherent silica for ultrasensitive detection of ovarian cancer in blood plasma.Pandey U, Rani MU, Deshpande AS, Singh SG, Agrawal A. ***Electrochimica Acta***. 397:139258, (**2021**). doi:10.1016/j.electacta.2021.139258

16. Antimicrobial surfaces: a review of synthetic approaches, applicability and outlook.Mahanta U, Khandelwal M, Deshpande AS. ***Journal of Materials Science***. 56(32):17915-17941, (**2021**). doi:10.1007/s10853-021-06404-0

17. N-doped MWCNTs from catalyst-free, direct pyrolysis of commercial glue.Damodar D, Mahanta U, Deshpande AS. ***Materials Chemistry and Physics***. 262:124319, (**2021**). doi:10.1016/j.matchemphys.2021.124319

18. In-situ formation of mesoporous SnO2@C nanocomposite electrode for supercapacitors.Rani MU, Naresh V, Damodar D, Muduli S, Martha SK, Deshpande AS. ***Electrochimica Acta***. 365:137284, (**2021**). doi:10.1016/j.electacta.2020.137284

19. Single-phase Gd0.2La0.2Ce0.2Hf0.2Zr0.2O2 and Gd0.2La0.2Y0.2Hf0.2Zr0.2O2 nanoparticles as efficient photocatalysts for the reduction of Cr(VI) and degradation of methylene blue dye.Anandkumar M, Lathe A, Palve AM, Deshpande AS. ***Journal of Alloys and Compounds***. 850:156716, (**2021**). doi:10.1016/j.jallcom.2020.156716

20. Structural and luminescent properties of Eu3+ doped multi-principal component Ce0.2Gd0.2Hf0.2La0.2Zr0.2O2 nanoparticles.Anandkumar M, Bagul PM, Deshpande AS. ***Journal of Alloys and Compounds***. 838:155595, (**2020**). doi:10.1016/j.jallcom.2020.155595

21. Wood-Derived Carbon Fibers Embedded with SnO x Nanoparticles as Anode Material for Lithium-Ion Batteries.Revathi J, Jyothirmayi A, Rao TN, Deshpande AS. ***Global Challenges***. 4(1):1900048, (**2020**). doi:10.1002/gch2.201900048

22. Corn husk derived activated carbon with enhanced electrochemical performance for high-voltage supercapacitors.Usha Rani M, Nanaji K, Rao TN, Deshpande AS. ***Journal of Power Sources***. 471:228387, (**2020**). doi:10.1016/j.jpowsour.2020.228387

23. Hard carbon derived from sepals of Palmyra palm fruit calyx as an anode for sodium-ion batteries.Damodar D, Ghosh S, Usha Rani M, Martha SK, Deshpande AS. ***Journal of Power Sources***. 438:227008, (**2019**). doi:10.1016/j.jpowsour.2019.227008

24. Wetting Transition from Lotus Leaf to Rose Petal using Modified Fly Ash.Mahanta U, Khandelwal M, Deshpande AS. ***ChemistrySelect***. 4(27):7936-7942, (**2019**). doi:10.1002/slct.201901535

25. 3D printable SiO 2 nanoparticle ink for patient specific bone regeneration.Roopavath UK, Soni R, Mahanta U, Deshpande AS, Rath SN. ***RSC Advances***. 9(41):23832-23842, (**2019**). doi:10.1039/C9RA03641E

26. Low temperature synthesis and characterization of single phase multi-component fluorite oxide nanoparticle sols.Anandkumar M, Bhattacharya S, Deshpande AS. ***RSC Advances***. 9(46):26825-26830, (**2019**). doi:10.1039/C9RA04636D

27. Near-Room-Temperature Synthesis of Sulfonated Carbon Nanoplates and Their Catalytic Application.Damodar D, Kunamalla A, Varkolu M, Maity SK, Deshpande AS. ***ACS Sustainable Chemistry & Engineering***. 7(15):12707-12717, (**2019**). doi:10.1021/acssuschemeng.8b06280

28. Sodium alginate/gelatin with silica nanoparticles a novel hydrogel for 3D printing.Soni R, Roopavath UK, Mahanta U, Deshpande AS, Rath SN. In: Sahulhameedu S. Shakya S. CJ, ed. *AIP Conference Proceedings*. Vol 1966. American Institute of Physics Inc.; 2018:020002. doi:10.1063/1.5038681

29. Nitrogen-doped graphene-like carbon nanosheets from commercial glue: morphology, phase evolution and Li-ion battery performance.Damodar D, Kumar SK, Martha SK, Deshpande AS. ***Dalton Transactions***. 47(35):12218-12227, (**2018**). doi:10.1039/C8DT01787E

30. Primary Structure and Phosphorylation of Dentin Matrix Protein 1 (DMP1) and Dentin Phosphophoryn (DPP) Uniquely Determine Their Role in Biomineralization.Deshpande AS, Fang P-A, Zhang X, Jayaraman T, Sfeir C, Beniash E. ***Biomacromolecules***. 12(8):2933-2945, (**2011**). doi:10.1021/bm2005214

31. Possible role of DMP1 in dentin mineralization.Beniash E, Deshpande AS, Fang PA, Lieb NS, Zhang X, Sfeir CS. ***Journal of Structural Biology***. 174(1):100-106, (**2011**). doi:10.1016/j.jsb.2010.11.013

32. Amelogenin-Collagen Interactions Regulate Calcium Phosphate Mineralization in Vitro.Deshpande AS, Fang P-A, Simmer JP, Margolis HC, Beniash E. ***Journal of Biological Chemistry***. 285(25):19277-19287, (**2010**). doi:10.1074/jbc.M109.079939

33. Bioinspired Synthesis of Mineralized Collagen Fibrils.Deshpande AS, Beniash E. ***Crystal Growth & Design***. 8(8):3084-3090, (**2008**). doi:10.1021/cg800252f

34. Synthesis of mesoporous ceria zirconia beads.Deshpande AS, Niederberger M. ***Microporous and Mesoporous Materials***. 101(3):413-418, (**2007**). doi:10.1016/j.micromeso.2006.11.036

35. Atomic-scale structure of nanocrystalline CeO 2 –ZrO 2 oxides by total x-ray diffraction and pair distribution function analysis.Gateshki M, Niederberger M, Deshpande AS, Ren Y, Petkov V. ***Journal of Physics: Condensed Matter***. 19(15):156205, (**2007**). doi:10.1088/0953-8984/19/15/156205

36. Hierarchically Structured Ceramics by High-Precision Nanoparticle Casting of Wood.Deshpande AS, Burgert I, Paris O. ***Small***. 2(8-9):994-998, (**2006**). doi:10.1002/smll.200600203

37. Steam reforming of methanol over Cu/ZrO/CeO catalysts: a kinetic study.MASTALIR A, FRANK B, SZIZYBALSKI A, et al. ***Journal of Catalysis***. 230(2):464-475, (**2005**). doi:10.1016/j.jcat.2004.12.020

38. Titania and Mixed Titania/Aluminum, Gallium, or Indium Oxide Spheres: Sol-Gel/Template Synthesis and Photocatalytic Properties.Deshpande AS, Shchukin DG, Ustinovich E, Antonietti M, Caruso RA. ***Advanced Functional Materials***. 15(2):239-245, (**2005**). doi:10.1002/adfm.200400220

39. Controlled Assembly of Preformed Ceria Nanocrystals into Highly Ordered 3D Nanostructures.Deshpande A, Pinna N, Smarsly B, Antonietti M, Niederberger M. ***Small***. 1(3):313-316, (**2005**). doi:10.1002/smll.200400060

40. Synthesis and Characterization of Stable and Crystalline Ce 1- x Zr x O 2 Nanoparticle Sols.Deshpande AS, Pinna N, Beato P, Antonietti M, Niederberger M. ***Chemistry of Materials***. 16(13):2599-2604, (**2004**). doi:10.1021/cm040155w

41. Synthesis of nanosized Ce0.75Zr0.25O2 porous powders via an autoignition: glycine nitrate process.Potdar H., Deshpande S., Khollam Y., Deshpande A., Date S. ***Materials Letters***. 57(5-6):1066-1071, (**2003**). doi:10.1016/S0167-577X(02)00932-1

42. A self-sustaining acid-base reaction in semi-aqueous media for synthesis of barium titanyl oxalate leading to BaTiO3 powders.Khollam Y., Deshpande A., Potdar H., Deshpande S., Date S., Patil A. ***Materials Letters***. 55(3):175-181, (**2002**). doi:10.1016/S0167-577X(01)00642-5

43. Preparation of ceria–zirconia (Ce0.75Zr0.25O2) powders by microwave–hydrothermal (MH) route.Potdar HS, Deshpande SB, Deshpande AS, et al. ***Materials Chemistry and Physics***. 74(3):306-312, (**2002**). doi:10.1016/S0254-0584(01)00485-0

44. Synthesis of yttria stabilized cubic zirconia (YSZ) powders by microwave-hydrothermal route.Khollam YB, Deshpande AS, Patil AJ, Potdar HS, Deshpande SB, Date SK. ***Materials Chemistry and Physics***. 71(3):235-241, (**2001**). doi:10.1016/S0254-0584(01)00287-5

45. Improved chemical route for quantitative precipitation of lead zirconyl oxalate (PZO) leading to lead zirconate (PZ) powders.Deshpande AS, Khollam YB, Patil AJ, Deshpande

SB, Potdar HS, Date SK. ***Materials Letters***. 51(2):161-171, (**2001**). doi:10.1016/S0167-577X(01)00284-1

46. Microwave-hydrothermal synthesis of equi-axed and submicron-sized BaTiO3 powders.Khollam YB, Deshpande AS, Patil AJ, Potdar HS, Deshpande SB, Date SK. ***Materials Chemistry and Physics***. 71(3):304-308, (**2001**). doi:10.1016/S0254-0584(01)00286-3

47. Simplified chemical route for the synthesis of barium titanyl oxalate (BTO).Potdar HS, Deshpande SB, Deshpande AS, et al. ***International Journal of Inorganic Materials***. 3(7):613-623, (**2001**). doi:10.1016/S1466-6049(01)00168-4

48. Preparation and characterization of strontium zirconate (SrZrO3) fine powders.Potdar HS, Deshpande SB, Patil AJ, Deshpande AS, Khollam YB, Date SK. ***Materials Chemistry and Physics***. 65(2):178-185, (**2000**). doi:10.1016/S0254-0584(00)00238-8

Patents:

1. Improved process for Wood derived Carbon - Metal oxide composites prepared by nanocasting of wood for electrode materials in lithium ion batteries, Janardhanan Revathi, **Atul Suresh Deshpande**, Tata Narasinga Rao, **Indian Pat. granted (2021) 20161103453**1

2. An improved process for the preparation of stable nano silver suspension having antimicrobial activity, Janardhanan Revathi, Nellipudi Satya Moulika, Avvaru Venkata Sai, **Atul Suresh Deshpande**, Karuppiah Murugan, Neha Yeshwanta Hebalkar, Ravula Vijay, Tata Narasinga Rao, Govindan Sundararajan, **Indian Pat. Appl.(2016) 201611027145**

3. An improved process for the preparation of barium titanyl oxalate (BTO)., Potdar, H.S., Deshpande, S.B., Date, S.K., Khollam**,** Y.B., **Deshpande**, **A.S.,** Patil, A.J., **Indian Pat. Appli.792/DEL/2001, Patent No. 220791.**

Recent Talks and conference proceedings

1. **Atul Suresh Deshpande,** Damodar D, M. Usha rani, "Precursor dictated morphology control of nanostructured carbons" (Oral Presentation) **Carbon MEMS: New Horizons, Fourth International Carbon-MEMS Meeting, Hyderabad, December 2018**

2. Atul Suresh Deshpande, M. Anandkumar, Saswata Bhattachrya, Ranjith Ramadurai, "Entropy Stabilized Multicomponent Oxides: Synthesis and Structural Analysis" (Invited talk) **NMD-ATM 2016, IIT Kanpur, November 2016**

3.M. Anandkumar, **Atul Suresh Deshpande**, Saswata Bhattachrya, Ranjith Ramadurai, "Entropy stabilized rare-earth based oxide: Synthesis and Thermal Stability, (Oral presentation) **MRS fall meeting, November 2016**

4. Atul Suresh Deshpande* High Entropy Oxides: Synthesis, Phase Stability, Thermal Stability and Functional Behavior (invited talk) International Conference on Powder Metallurgy and Particulate Materials **(PM 23)**, Mumbai **(2023)**

5. E. Majhi, **Atul Suresh Deshpande*** Okra-Derived Hierarchical Porous Structured Activated Carbon Tuning with Metal-Oxide for Advanced Supercapacitors (oral presentation) **MRS fall meeting, November 2023**

6. Atul Suresh Deshpande* Engineering the properties of carbon to achieve high energy density and power density simultaneously for supercapacitors. (Invited talk) International Conference on Powder Metallurgy and Particulate Materials **(PM 25)**, Mumbai **(2025)**